

PETER TRAN

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Australian Citizen · Security Clearance: Negative Vetting 1

EDUCATION

University of Queensland

Bachelor of Engineering (Honours) & Master of Engineering (Software)
First Class Honours – GPA: 6.35/7

Brisbane, Australia
Feb 2020 - Dec 2024

WORK EXPERIENCE

Commonwealth Bank of Australia

Backend Software Engineer

Sydney, Australia
Feb 2026 - Present

NetBank Next – rebuilding the backend of CommBank’s online banking (17M+ customers, 170 TPS) onto a federated GraphQL API – one of the bank’s first large-scale GraphQL federation deployments

- Drove architectural decisions for the GraphQL federation platform – supergraph composition, schema governance, observability, subgraph CI/CD – and implemented its core infrastructure, enabling 10+ service domains to independently build and deploy subgraphs
- Built and own three GraphQL subgraphs (Accounts, Transactions, Parties), migrating legacy REST/SOAP endpoints onto the federated layer and cutting p95 latency from 500ms to 300ms (40%)
- Designed a custom GraphQL directive enforcing OAuth 2.0 scope-based authorization, giving field- and entity-level access control over sensitive financial data
- Built an MCP server exposing the supergraph to AI agents, used by 200+ engineers to query the schema and customer usage in natural language instead of hand-writing GraphQL introspection
- Serve as on-call engineer (PagerDuty) for the GraphQL platform serving 17M customers, operating against a 99.9% availability SLO

Graduate Software Engineer

Feb 2025 - Feb 2026

- Built event-driven ingestion and storage pipelines (SQS → Lambda → S3/PostgreSQL → Kafka) within CommBank’s Payment Data Platform, normalizing 5,000 payment events/sec into ISO 20022 as a common cross-bank, cross-country standard
- Joined NetBank Next at inception (Sep 2025), built its first subgraphs – promoted to the platform team

Boeing Australia

Graduate Software Engineer

Brisbane, Australia
Nov 2024 - Feb 2025

Undergraduate Software Engineer

Nov 2021 - Nov 2024

Safety-critical, distributed embedded systems engineering on the MQ-28 Ghost Bat, Boeing’s autonomous fighter aircraft

- Built sensor data ingestion software for the MQ-28 mission computer on Green Hills INTEGRITY RTOS, meeting DO-178C safety-critical certification standards
- Developed C++ scenario-based testing frameworks and hardware-in-the-loop (HWIL) infrastructure (Python, MATLAB), verifying mission computer software against real avionics hardware
- Built a custom internal React dashboard for monitoring health and gathering metrics from air-gapped physical components and servers in the aircraft's digital twin, cutting hardware fault diagnosis time from 2 hours to 2 minutes
- Completed a master's thesis with Boeing evaluating Rust as a certifiable replacement for C++ and Ada in DO-178C safety-critical aviation software

Services Australia

Software Engineering Intern

Brisbane, Australia
Feb 2022 - Nov 2023

- Optimised the myGov Inbox, a React-based messaging feature serving millions of citizens across myGov, Centrelink, and Medicare, improving page load times by 30%

SKILLS

Languages:	TypeScript, Java, Python, C/C++, Rust
Frameworks & APIs:	GraphQL (Apollo Federation), Spring Boot, React, Node.js, OAuth 2.0
Infrastructure & Tools:	AWS, DynamoDB, PostgreSQL, Kafka, Docker, GitHub Actions, Grafana
Certifications:	AWS Certified Developer – Associate

PROJECTS

Fit2Track Node.js, React, React Native, PostgreSQL, AWS, LangChain

A fitness app where users create challenges triggered by QR codes or geofencing. Features a WYSIWYG editor, AI-generated content, Google Maps, and Google Sign-In; deployed on auto-scaling EC2 + ALB.

Retrieval-Augmented Generation Pipeline Python, PyTorch, Machine Learning

Built a RAG pipeline with Hugging Face Transformers and FAISS, benchmarking BM25, DPR, and TILDEv2 retrieval; TILDEv2 outperformed BM25 (nDCG@3: 0.77 vs. 0.48, +61%) with statistical significance.

PROGRAMS & TRAINING

Australian Defence Force Cyber GAP Program

Australian Defence Force

Selected for a competitive 12-month ADF cybersecurity development program; completed hands-on training in digital forensics, threat and risk assessment, and cybersecurity fundamentals

Jan 2022 - Dec 2022